# CHAPTER-8
# SIMULATION RESULTS

## 8.1 Schematic Diagrams:

### 8.1.1 Existing 10T SRAM Cell Schematic Circuit:

Using Tanner EDA, the current 10 T SRAM cell was designed and suggested as 10 T SRAM cell. The conventional 10 T SRAM cell is built using 32nm MOS technology and uses Tanner EDA to develop a cell design. Figure 8.3 shows the current and planned circuit schematics for a 10 T SRAM cell. An analysis of the features of the product's performance characteristics, such as power, delay, and power delay, is done. 6T SRAM cell with 4 additional transistors is essentially the same as the traditional 10T SRAM cell.To prevent the flipping of the cell, an extra read circuitry is added

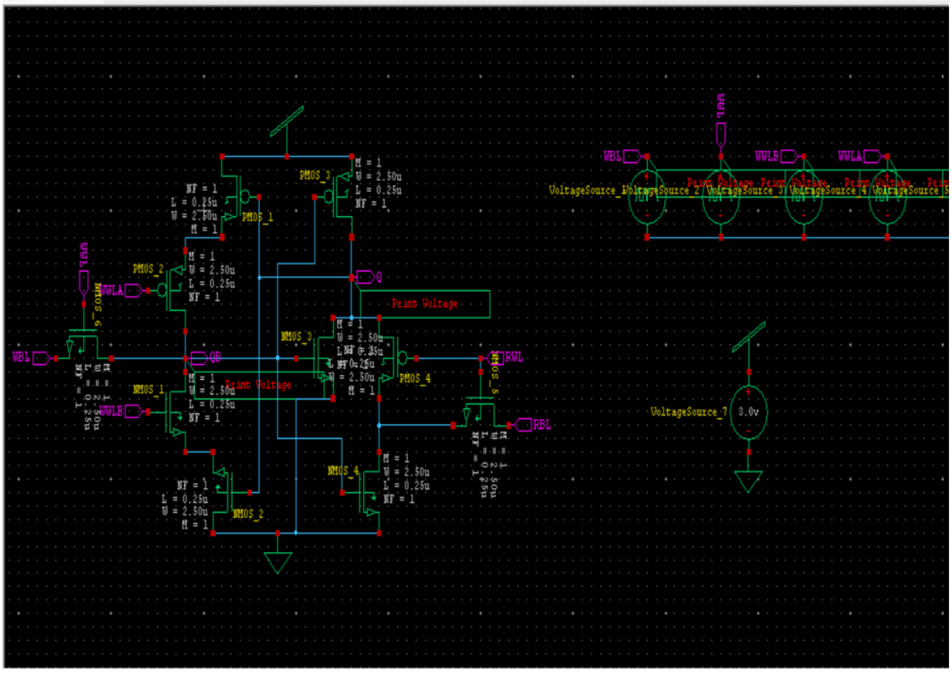

**Fig 8.1: Schematic Circuit of Existing 10T SRAM Cell**

## Inference:

For a Write operation in the memory cell, BL is set to 1, while the BLB is set to 0. As a result, the output Q represents the write data to the memory cell, which is set to 1, while the complementary output QB represents the read data from the memory cell and is set to 0.

### 8.1.2 Proposed 10T SRAM Cell Schematic Circuit:

Tanner EDA is used to design a 10 T SRAM cell at 18nm using MOS technology. The suggested cell's schematic circuit is shown in Figure 8.6. power, delay, and power delay product are examined; performance parameters such as power, delay, and power delay are investigated Table II presents the circuit assessment.

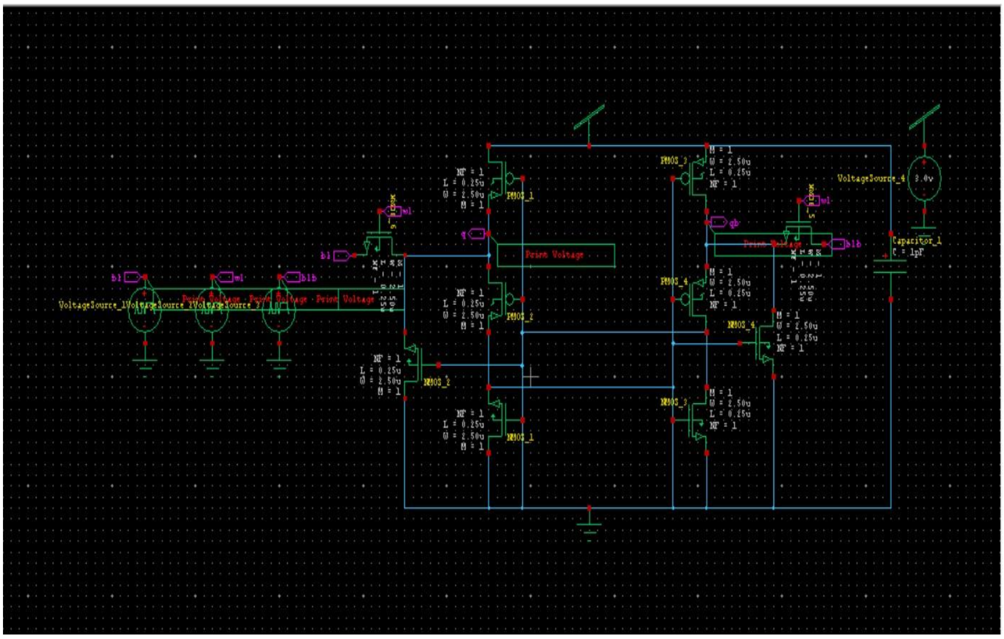

**Fig 8.2: Schematic Circuit of Proposed 10T SRAM Cell**

### Inference:

For a read operation in the memory cell, the blb is set to 1, while the bl is set to 0. As a result, the output qb represents the read data from the memory cell, which is set to1, while the complementary output q represents the write data to the  memory cell and is set to 0.

## 8.2 Output Waveforms:

### 8.2.1 Existing 10T SRAM Cell:

The Fig.8.3 shows the simulation results of traditional 10 T SRAM cell using Tanner EDA tool. The power consumption of existing 10 T SRAM cell with 32nm techonology is 0.29mW and the overall time accessing is 1.19 sec.

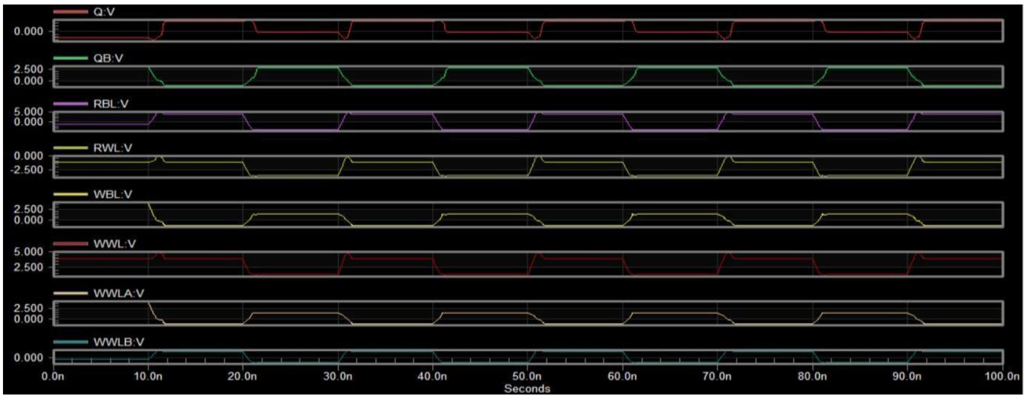


**Fig 8.3: Output Waveforms of Existing 10T SRAM Cell**

## Inference:

For a read operation in the memory cell, the Read Bit Line (RBL) is set to 1, while the Write Bit Line (WBL) is set to 0. As a result, the output Q represents the read data from the memory cell, which is set to 1, while the complementary output QB represents the write data to the memory cell and is set to 0.

## 8.2.2 Proposed 10T SRAM Cell:

The Fig.8.4 shows the simulation results of Proposed 10 T SRAM cell using Tanner EDA tool. The power consumption of proposed 10 T SRAM cell with 18nm technology is 0.17mW and the overall time accessing is 0.9sec.

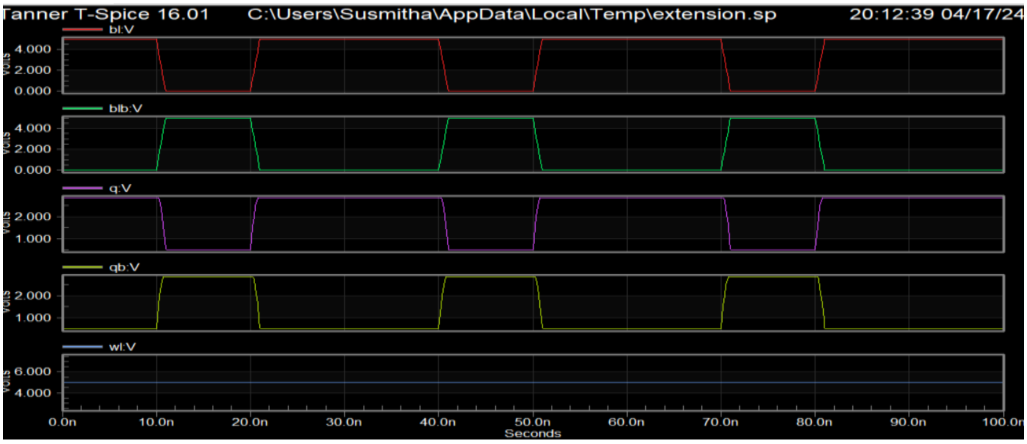


**Fig 8.4: Output Waveforms of Proposed 10T SRAM Cell**

## Inference:

For a Write operation in the memory cell, bl is set to 1, while the blb is set to 0. As a result, the output q represents the write data to the memory cell, which is set to 1, while the complementary o/p qb represents the read data from the memory cell and is set to 0.

## 8.3 Time Delay and Power Analysis8.3.1Existing 10T SRAM Cell:

### 8.3.1 Existed 10T SRAM Cell:

```
Power Results

VVoltageSource_7 from time 0 to 100
Average power consumed -> 1.626605e-013 watts
Max power 2.905879e-003 at time 1.15105e-008
Min power 2.873311e-011 at time 0


Parsing                        0.00 seconds
Setup                          0.01 seconds
DC operating point             0.06 seconds
Transient Analysis             0.04 seconds
Overhead                       1.07 seconds
-----------------------------------------
Total                          1.19 seconds
```

**Fig 8.5: Power and Timing Analysis of Existing 10T SRAM Cell**

### 8.3.2 Proposed 10T SRAM Cell:

```
Power Results

VVoltageSource_2 from time 0 to 100
Average power consumed -> 8.952503e-016 watts
Max power 1.737586e-004 at time 4.09191e-008
Min power 0.000000e+000 at time 0


Parsing                        0.01 seconds
Setup                          0.00 seconds
DC operating point             0.08 seconds
Transient Analysis             0.00 seconds
Overhead                       0.80 seconds
-----------------------------------------
Total                          0.90 seconds
```

**Fig 8.6: Power and Timing Analysis of Proposed 10T SRAM Cell**

**TABLE-1**
**Parameters for Existed 10T SRAM Cell**

| Power (mWatts) | Delay (seconds) | Power Delay Product (Watts Sec) |
|---|---|---|
| 0.29 | 1.19 | $3.4\times10^{-1}$ |

**TABLE-2**
**Parameters for Proposed 10T SRAM Cell**

| Power (mWatts) | Delay (seconds) | Power Delay Product (Watts Sec) |
|---|---|---|
| 0.17 | 0.9 | $1.5\times10^{-1}$ |

**TABLE-3**
**Comparison Table**

| Parameter | Existing 10T SRAM | Proposed 10T SRAM |
|---|---|---|
| **Number of Nodes** | 12 | 11 |
| **Power (mWatts)** | 0.29 | 0.17 |
| **Delay (seconds)** | 1.19 | 0.9 |
| **Power Delay Product (Watts sec)** | $3.4\times10^{-1}$ | $1.5\times10^{-1}$ |